IILM UNIVERSITY

EduSkills Foundation | AICTE Virtual Internship

VIRTUAL INTERNSHIP PRESENTATION

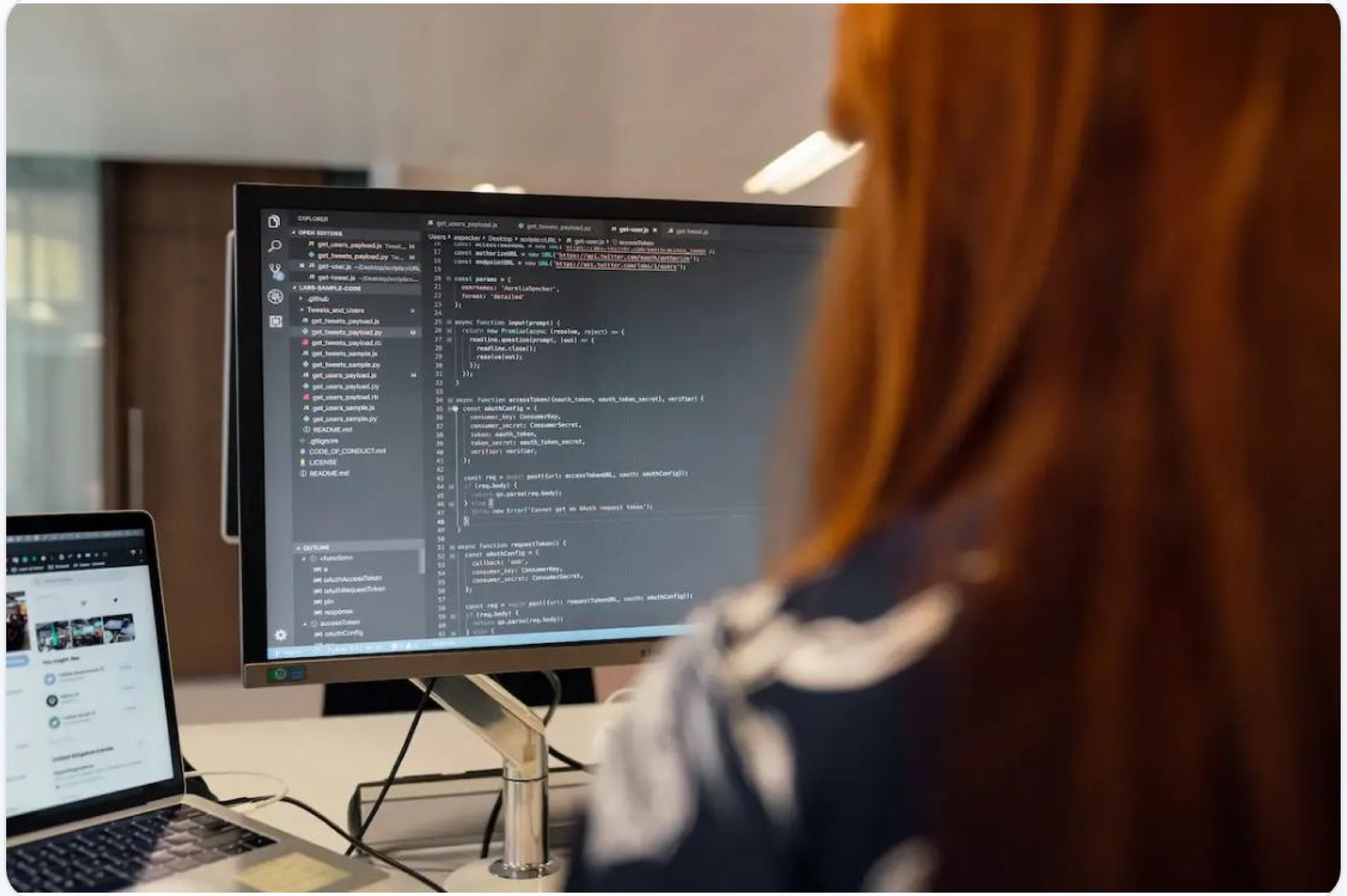
Java Full Stack Developer

A Comprehensive 10-Week Technical Internship Report & Showcase

STUDENT NAME	ENROLLMENT NUMBER	PROGRAM & SEMESTER	ORGANIZATION NAME
Nidhi Kumari	2410030430	B.Tech (CSE) - 5th Sem	EduSkills Foundation

Executive Summary

- **Program Scope:** Completed an intensive 10-week Virtual Internship Program in Java Full Stack Web Development from January 2026 to March 2026.
- **Patronage & Strategic Partner:** Executed by EduSkills Foundation under the patronage of AICTE (All India Council for Technical Education).
- **Core Domain Focus:** End-to-end web engineering covering modern HTML/CSS/JS frontend design, Core & Advanced Java programming, enterprise Spring Boot APIs, and Hibernate MySQL persistence.
- **Certification Credential:** Successfully passed all weekly progress milestones and final evaluation test with Certificate ID: **2026-E2CF1F0C43**.



Organization Profile

EduSkills Foundation

EduSkills is a premier non-profit social enterprise enabling Industry 4.0 digital workforce skills across higher education institutions in India. ISO 9001:2015 and ISO 20010 certified organization.

AICTE Support

Supported under the official AICTE National Internship Portal initiative, bridging the critical gap between academic curriculum and industrial software engineering practices.

Methodology

Features structured week-by-week learning modules, hands-on lab exercises, practical project submission, source control workflows, and comprehensive final evaluation.

Curriculum Roadmap

Week	Module Title	Core Competencies & Key Deliverables	Stack
Week 1	HTML Fundamentals	Page structures, semantic tags, accessibility, audio/video elements, tables & forms	Frontend
Week 2	CSS & Responsive Design	Selectors, box model, CSS Grid, Flexbox alignment, media queries, keyframe animations	Styling
Week 3	Bootstrap Framework	Responsive grid system, navbar, cards, modals, utilities, mobile-first design	UI Library
Week 4	JavaScript & jQuery	ES6 syntax, DOM manipulation, asynchronous AJAX events, dynamic web interfaces	Scripting
Week 5–6	Core & Advanced Java	OOP principles, Collections Framework, Exception handling, JDBC, Servlets, Multithreading	Backend
Week 7–8	Spring Framework & Spring Boot	Dependency Injection, REST API endpoints, ORM mapping, MySQL CRUD operations	Framework
Week 9–10	Git & Version Control	Source version control, GitHub team workflows, branching strategies, comprehensive exam	DevOps

Frontend Architecture & Stack

- **HTML5 Semantic Markup:** Constructed accessible document structures, forms, audio/video elements, and clean SEO web hierarchy.
- **CSS3 Styling & Layouts:** Implemented responsive design patterns using modern CSS Flexbox, Grid systems, and CSS variables.
- **Bootstrap 5 Framework:** Built mobile-first responsive components, interactive modal dialogs, and clean UI interfaces efficiently.

1. HTML5 Document Structure
Parses DOM tree & defines semantic web elements

↓

2. CSS3 & Bootstrap 5 Styling
Applies responsive Grid, Flexbox & mobile-first UI components

↓

3. JS & jQuery Dynamics
Executes DOM event handlers, AJAX queries & interactive UI logic

JavaScript & jQuery

ES6+ JavaScript

Leveraged modern JavaScript concepts including arrow functions, promises, async/await, destructured objects, and ES modules for efficient client-side logic execution.

DOM & Event Model

Mastered dynamic HTML Document Object Model (DOM) traversal, custom event handlers, form data validation, and browser storage (LocalStorage/SessionStorage).

jQuery & AJAX

Simplified client-side scripting and asynchronous HTTP calls using jQuery, enabling smooth page updates without requiring browser reloads.

Core Java Foundations

- **Object-Oriented Programming:** Practical implementation of Encapsulation, Inheritance, Polymorphism, and Abstraction in Java applications.
- **Java Collections Framework:** In-depth hands-on experience with Lists, Sets, Maps, Queues, and Iterator algorithms for high-performance memory data processing.
- **Exception & File I/O:** Robust error management using try-catch-finally blocks, custom exception classes, and BufferedReader/BufferedWriter file handling streams.

Advanced Java Engineering

Java Servlets

Developed server-side web controllers utilizing Java Servlets to parse HTTP request-response lifecycle objects dynamically across Java EE applications.

JDBC API

Established database connectivity via Java Database Connectivity (JDBC) API with Connection Pooling and PreparedStatement handling for secure queries.

Multithreading

Implemented multithreaded task execution strategies using the Java Concurrency Utilities and Executor Framework for high-throughput request processing.

Spring Boot Architecture

Dependency Injection

Utilized Inversion of Control (IoC) container and Spring Dependency Injection (@Autowired, @Component, @Service, @Repository) for loosely-coupled modular architecture.

Spring Boot & Microservices

Accelerated Enterprise application bootup using auto-configuration, embedded Tomcat server instances, and Spring Boot Starters.

REST API Development

Built robust HTTP RESTful web services with JSON payloads (@RestController, @GetMapping, @PostMapping) adhering to modern backend industry standards.

Data Persistence Architecture

- **Hibernate ORM Framework:** Seamlessly mapped Java domain objects (@Entity, @Table) to MySQL relational database tables using JPA annotations.
- **MySQL Relational Schema:** Designed normalized database tables, indexed primary/foreign keys, and executed automated CRUD operations.
- **Spring Data JPA Repositories:** Leveraged repository interfaces to perform transactional data persistence and query generation efficiently.

Spring Boot Application
Java Controllers & Service Layer (@Entity POJO)

↕ Object-Relational Mapping (JPA)

Hibernate ORM Engine
Session Management & HQL Query Generator

↕ JDBC Connection Driver

MySQL Database
Relational Tables & Transactional Storage

Git & Version Control

- **Distributed Version Control:** Mastered fundamental Git CLI commands (`git init`, `add`, `commit`, `push`, `pull`, `rebase`) for tracking local code history.
- **GitHub Collaboration:** Managed remote repositories, established Git feature-branching strategies, and raised Pull Requests (PRs) for code review.
- **Conflict Resolution:** Applied effective strategies to resolve merge conflicts and maintain clean branch commit logs during team software building.

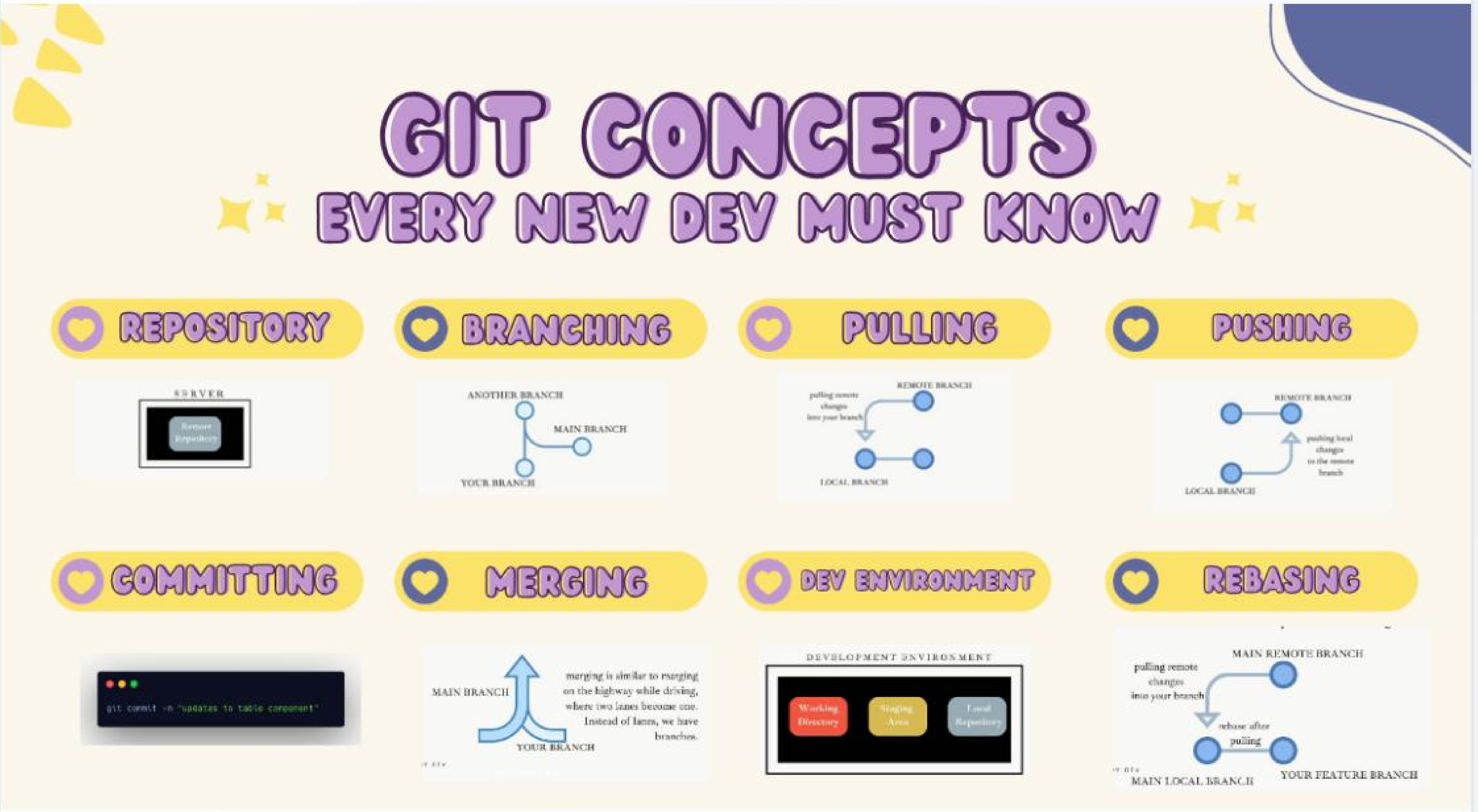
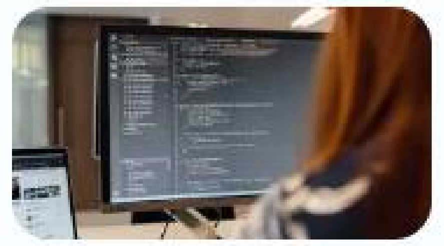


Image Sources



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